Performance Report: Interactive Image Mosaic Generator

1. Introduction

The Interactive Image Mosaic Generator is a tool that converts an uploaded image into a mosaic representation using different tile patterns and grid sizes. The goal is to create visually appealing mosaics and measure how closely the mosaic resembles the original image using quantitative metrics.

2. Methodology

2.1 Image Processing

The input image is resized to a consistent size of 512x512 pixels for uniform processing. The image is divided into grid cells based on the chosen grid size (ranging from 8x8 to 64x64). Each grid cell is replaced with a colored tile, where the color is determined by the average color of the corresponding cell in the original image.

2.2 Tile Design

Four different tile patterns are supported:

- **Solid Tile**: A single-color tile based on the average color of the cell.
- **Checker Tile**: A checkerboard pattern of the average color.
- **Circles Tile**: A circular pattern with the average color as the fill.
- **Diagonal Tile**: Diagonal lines using the average color.

2.3 Similarity Metrics

Two performance metrics are used to evaluate the similarity between the original image and the reconstructed mosaic:

- **Mean Squared Error (MSE)**: Measures the average squared difference between the original and mosaic images. A lower MSE indicates higher similarity.
- **Structural Similarity Index (SSIM)**: A perceptual metric that quantifies image quality degradation. The SSIM score ranges from 0 to 1, with values closer to 1 indicating higher similarity.

3. Implementation

The project is implemented using **Python** with the following libraries:

- **OpenCV** for image manipulation.
- **Gradio** for building the interactive web interface.
- **NumPy** for numerical operations.
- **scikit-image** for SSIM calculation.
- **PIL (Pillow)** for image processing.

User Interface

The Gradio interface allows users to:

1. Upload an image.
2. Select grid size and tile pattern.
3. View the original image, grid segmentation preview, and mosaic output.
4. Observe the calculated similarity metrics (SSIM and MSE).

4. Results and Discussion

4.1 Performance

- **SSIM** and **MSE** provide a reliable measure of mosaic quality. SSIM focuses on structural similarity, while MSE quantifies pixel-level differences.
- Increasing the grid size reduces the tile size, resulting in a more detailed mosaic but potentially higher computational cost.
- Pattern selection impacts the mosaic's visual appearance, offering creative flexibility.

4.2 Challenges

- Handling edge cases where grid cells at image boundaries may be smaller than the specified tile size.
- Balancing computational efficiency with visual quality, especially for larger grid sizes.

4.3 Optimizations

- Implementing multi-threading for faster tile generation.
- Allowing users to customize additional patterns and adjust color blending for smoother transitions.

5. Conclusion

The Interactive Image Mosaic Generator successfully creates mosaics with customizable patterns and grid sizes. The use of SSIM and MSE as performance metrics ensures objective evaluation of mosaic quality. The tool provides an intuitive interface for users to explore image processing creatively.

6. References

- scikit-image Documentation: https://scikit-image.org/
- Gradio Documentation: https://gradio.app/
- OpenCV Documentation: https://opencv.org/
